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ME 5130: Advanced Heat and Mass Transfer

Homework 01: Heat Conduction

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1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \quad (1)$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta) \quad (2)$$

1–2, Solution: